STAT 618
Bayesian Statistics

Quiz # 6

April 11, 2018

Name: SOLUTION

Let U be a standard uniform random variable. Derive a formula that will convert U into a Pareto$(2,5)$ random variable X with the density $f(x) = 50x^{-3}$ and the cumulative distribution function $F(x) = 1 - \left(\frac{5}{x}\right)^2$, $x \geq 5$.

Compute X using your algorithm if our computer generated $U = 0.36$

SOLUTION. We'll use the inverse transform algorithm, $X = F^{-1}(U)$. To find X, solve the equation $F(X) = U$ for X,

$$1 - \left(\frac{5}{X}\right)^2 = U \quad \Leftrightarrow \quad \left(\frac{5}{X}\right)^2 = 1 - U \quad \Leftrightarrow \quad \frac{5}{X} = \sqrt{1-U} \quad \Leftrightarrow \quad X = \frac{5}{\sqrt{1-U}}.$$

For $U = 0.36$, $X = \dfrac{5}{\sqrt{1-0.36}} = \dfrac{5}{0.8} = 6.25$.

Since $(1-U)$ is also a standard uniform random variable, the answer $X = \dfrac{5}{\sqrt{U}} = \dfrac{5}{\sqrt{0.36}} = 8.33$ is just as good.

It's useful to check that both values of X belong to the range of possible values of this Pareto distribution, $X \geq 5$.

Inventory of some distributions

Discrete distributions					
Distribution	Parameters	Probability mass function	$\mathbf{E}(X)$	$\mathrm{Var}(X)$	Comments
Binomial $\mathcal{B}(n,p)$	$n=1,2,\ldots$ $p \in (0,1)$	$\binom{n}{x} p^x (1-p)^{n-x}$, $x=0,1,\ldots,n$	np	$np(1-p)$	
Geometric $\mathcal{G}e(p)$	$p \in (0,1)$	$(1-p)^{x-1}p,\ x=1,2,\ldots$	$1/p$	$(1-p)/p^2$	$\mathcal{NB}(1,p)$
Negative Binomial $\mathcal{NB}(k,p)$	$k=1,2,\ldots$ $p \in (0,1)$	$\binom{x-1}{r-1} p^k (1-p)^{x-k}$, $x=k,k+1,\ldots$	$1/p$	$(1-p)/p^2$	
Poisson $\mathcal{P}(\lambda)$	$\lambda > 0$	$e^{-\lambda}\lambda^x/x!,\ x=0,1,2,\ldots$	λ	λ	

Continuous distributions					
Distribution	Parameters	Density	$\mathbf{E}(X)$	$\mathrm{Var}(X)$	Comments
Normal $\mathcal{N}(\mu,\sigma)$	$\mu \in R$ $\sigma > 0$	$\frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{(x-\mu)^2}{2\sigma^2}},\ x \in R$	μ	σ^2	
Uniform $\mathcal{U}(a,b)$	$a < b$	$\frac{1}{b-a},\ a < x < b$	$\frac{a+b}{2}$	$\frac{(b-a)^2}{12}$	$Be(1,1)$
Gamma $\mathcal{G}(\alpha,\beta)$	$\alpha,\beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha} x^{\alpha-1} e^{-x/\beta}$ $x > 0$	$\alpha\beta$	$\alpha\beta^2$	
Exponential $\mathcal{E}(\beta)$	$\beta > 0$	$\frac{1}{\beta} e^{-x/\beta},\ x > 0$	β	β^2	$\mathcal{G}(1,\beta)$
Chi-square $\chi^2(\nu)$	$\nu > 0$ (d.f.)	$\frac{1}{2^{\nu/2}\Gamma(n/2)} x^{\nu/2-1} e^{-x/2}$ $x > 0$	ν	2ν	$\mathcal{G}(\nu/2,2)$
Beta $\mathcal{B}e(\alpha,\beta)$	$\alpha,\beta > 0$	$\frac{\Gamma(\alpha,\beta)}{\Gamma(\alpha)\Gamma(\beta)} x^{\alpha-1}(1-x)^{\beta-1}$ $0 < x < 1$	$\frac{\alpha}{\alpha+\beta}$	$\frac{\alpha\beta}{(\alpha+\beta)^2(\alpha+\beta+1)}$	
Cauchy $\mathcal{C}(\alpha,\beta)$	$\alpha \in R$ $\beta > 0$	$\frac{\beta}{\pi[\beta^2+(x-\alpha)^2]}, x \in R$	–	–	
Inverse Gamma $\mathcal{IG}(\alpha,\beta)$	$\alpha,\beta > 0$	$\frac{1}{\Gamma(\alpha)\beta^\alpha x^{\alpha+1}} e^{-1/x\beta}$ $x > 0$	$\frac{1}{\beta(\alpha-1)}$ if $\alpha > 1$	$\frac{1}{\beta^2(\alpha-1)^2(\alpha-2)}$ if $\alpha > 2$	
Pareto $\mathcal{P}a(x_0,\alpha)$	$\alpha, x_0 > 0$	$\frac{\alpha x_0^\alpha}{x^{\alpha+1}},\ x > x_0$	$\frac{\alpha x_0}{\alpha-1}$ if $\alpha > 1$	$\frac{\alpha x_0^2}{(\alpha-1)^2(\alpha-2)}$ if $\alpha > 2$	